

Grid-Based Optimization Mapping of a Rankine Cycle

An Interactive Thermodynamic Design Tool

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July 11, 2026

Abstract

This report presents the development of an interactive Rankine cycle optimization tool. The tool evaluates a grid of operating conditions by performing two-variable parametric sweeps, calculating cycle performance at each grid point, and visualizing the feasible design space. Engineering constraints, including minimum turbine outlet quality, minimum net work output, maximum heat input, and minimum thermal efficiency, can be applied to separate feasible and infeasible operating regions. The program then identifies the optimum design point according to a user-defined objective function, such as maximizing thermal efficiency, maximizing net work, minimizing heat input, or maximizing turbine outlet quality. The developed tool is intended to support the interpretation of thermodynamic trade-offs in simple vapor power cycles.

Contents

1	Introduction	3
2	Thermodynamic Model	3
2.1	Assumptions	3
2.2	State Points	3
2.3	Performance Parameters	4
3	Software Implementation	4
4	Optimization Methodology	5
5	Graphical User Interface	6
6	Results and Discussion	7
6.1	Effect of Boiler Pressure and Turbine Inlet Temperature	7
6.2	Effect of Condenser Pressure and Boiler Pressure	9
6.3	Objective Function Comparison	14

7	Conclusion and Further Developments	16
7.1	Conclusion	16
7.2	Further Developments	16

1 Introduction

Rankine cycles are widely used in vapor power systems and form the basis of many steam power plants. In a practical design problem, the best operating point is not determined only by maximizing thermal efficiency. Other factors such as turbine outlet quality, net work output, and heat input requirement must also be considered. For instance, an operating point with high efficiency may require a heat input that exceeds the desired design limit, while another point may maximize net work output at the expense of turbine outlet quality. Low outlet quality indicates a higher moisture content at the turbine exit, which can increase the risk of blade erosion and reduce the reliability of the turbine. For this reason, practical constraints such as maximum heat input, minimum net work output, and minimum turbine outlet quality are necessary when selecting a feasible design point. Therefore, the optimization of a Rankine cycle is not only a thermodynamic performance problem, but also a constrained engineering design problem.

2 Thermodynamic Model

The simple Rankine cycle considered in this project consists of four main processes. Process 1–2 represents the compression of saturated liquid water in the pump. For the ideal case, this process is treated as isentropic compression; however, pump irreversibility is included through the pump isentropic efficiency. Process 2–3 represents constant-pressure heat addition in the boiler, where the working fluid is heated to the superheated vapor region. Process 3–4 represents expansion through the turbine. The ideal turbine expansion is assumed to be isentropic, while the actual turbine outlet state is calculated using the turbine isentropic efficiency. Finally, process 4–1 represents constant-pressure heat rejection in the condenser, returning the working fluid to the saturated liquid state.

2.1 Assumptions

The Rankine cycle model used in this study is based on the following assumptions:

- The cycle operates at steady state.
- Changes in kinetic and potential energies are neglected.
- The pump and turbine are treated as adiabatic devices.
- The boiler and condenser operate at constant pressure.
- Pressure losses in the boiler, condenser, and connecting pipes are neglected.
- The working fluid is water.
- State 1 is assumed to be saturated liquid at the condenser pressure.
- State 3 is assumed to be superheated vapor at the turbine inlet.
- Pump and turbine irreversibilities are represented using isentropic efficiencies.

2.2 State Points

The four principal states of the cycle are defined as follows:

- State 1: Saturated liquid at the condenser outlet
- State 2: Compressed liquid at the boiler inlet
- State 3: Superheated vapor at the turbine inlet
- State 4: Turbine outlet state

2.3 Performance Parameters

The pump work is calculated using the energy balance as

$$w_{pump} = h_2 - h_1 \quad (1)$$

The turbine work is calculated as

$$w_{turbine} = h_3 - h_4 \quad (2)$$

The net work output is

$$w_{net} = w_{turbine} - w_{pump} \quad (3)$$

The heat input is

$$q_{in} = h_3 - h_2 \quad (4)$$

The thermal efficiency of the cycle is then given by

$$\eta_{th} = \frac{w_{net}}{q_{in}} \quad (5)$$

It can be observed that the main performance parameters of the cycle are directly related to the enthalpy values at the four state points. Therefore, the evaluation of thermodynamic state properties is a central part of the cycle analysis. All work and heat transfer quantities are expressed on a specific basis per unit mass of working fluid. The computational implementation of this property evaluation is discussed in the following section.

3 Software Implementation

The computational tool was developed in Python and consists of two main parts: a backend calculation module and an interactive graphical user interface. The backend module performs the thermodynamic calculations, parametric sweeps, constraint checks, and optimum point selection. The graphical interface allows the user to define operating ranges, component efficiencies, constraints, and the optimization objective.

Thermodynamic properties of water were evaluated using the CoolProp library, which provides the required state properties such as enthalpy, entropy, temperature, density, and

vapor quality. These properties were then used to solve the Rankine cycle automatically at each point in the selected design grid.

For each sweep type, the program creates a grid of operating conditions and calculates the corresponding performance parameters. The results are visualized as an optimization map, where thermal efficiency is shown as a contour plot, infeasible regions are separated according to the selected constraints, and the optimum valid design point is marked and summarized through the graphical interface.

4 Optimization Methodology

The program performs a two-variable parametric sweep over selected operating variables. For each point in the design space, the Rankine cycle is solved and the main performance quantities are calculated. In this way, the program evaluates a grid of operating conditions rather than a single design point.

Three sweep modes are available in the developed tool:

- **Boiler pressure vs. turbine inlet temperature:** the boiler pressure and turbine inlet temperature are varied while the condenser pressure is kept constant.
- **Condenser pressure vs. turbine inlet temperature:** the condenser pressure and turbine inlet temperature are varied while the boiler pressure is kept constant.
- **Condenser pressure vs. boiler pressure:** the condenser pressure and boiler pressure are varied while the turbine inlet temperature is kept constant.

For each sweep mode, the user defines the start, end, and step values of the selected operating variables. The program then calculates the cycle performance at every grid point and uses the resulting data to construct the optimization map.

The available objective functions are:

- Maximize thermal efficiency
- Maximize net work
- Minimize heat input
- Maximize turbine outlet quality

The program also applies user-defined engineering constraints. These constraints include minimum turbine outlet quality, minimum net work output, maximum heat input, and minimum thermal efficiency. Designs that violate one or more of these constraints are marked as infeasible and are excluded from the optimum point selection.

As a result, the selected optimum point is not simply the best point in the entire sweep domain, but the best feasible point according to the selected objective function and engineering constraints.

5 Graphical User Interface

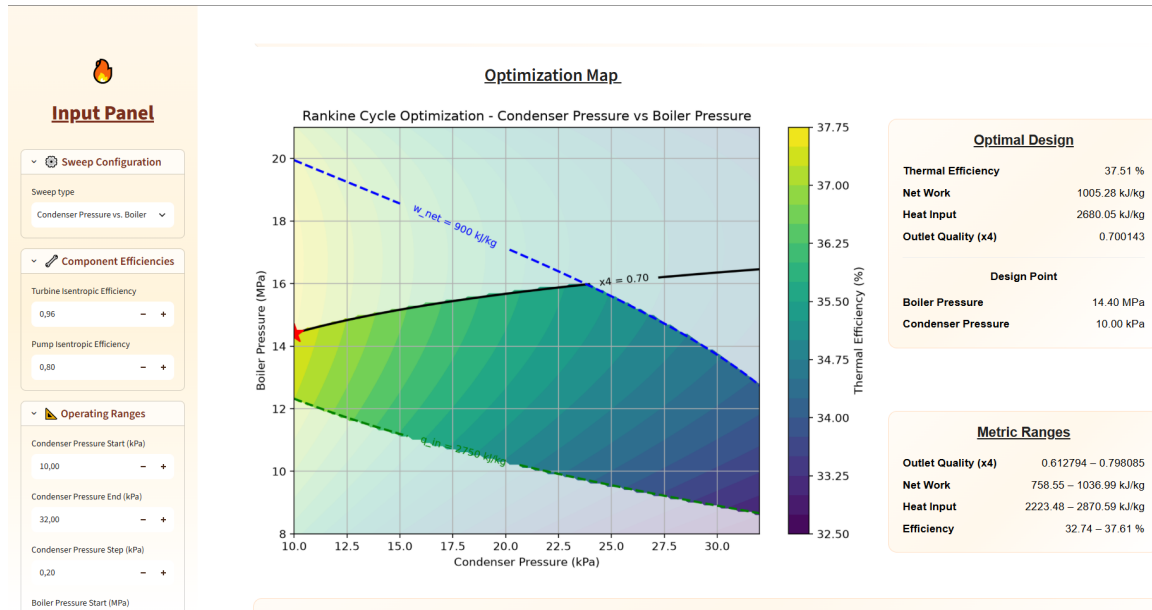


Figure 5.1: Graphical user interface of the Rankine cycle optimization tool showing the input panel, optimization map, optimum design point, and calculated metric ranges.

The graphical user interface was designed to make the Rankine cycle optimization process interactive and easy to interpret. The left side of the interface contains the **Input Panel**, where the user selects the sweep type, defines the operating ranges, sets the component efficiencies, imposes engineering constraints, and chooses the optimization objective. After defining these inputs, the user starts the calculation by pressing the **Run Analysis** button.

After the analysis is executed, the main section of the interface displays an optimization map showing the thermal efficiency distribution over the selected design space. Constraint boundaries are also shown on the map, making it possible to identify the feasible operating region. This region corresponds to the part of the design space that satisfies all selected engineering constraints, such as minimum turbine outlet quality, minimum net work output, maximum heat input, and minimum thermal efficiency.

The optimum feasible design point is marked on the map with a **red point**. This point represents the best valid operating condition according to the selected objective function. For example, if the objective is to maximize thermal efficiency, the red point indicates the feasible grid point with the highest thermal efficiency.

In addition to the optimization map, the interface displays summary cards for the optimum design and the calculated metric ranges. The **Optimum Design** card lists the main performance values at the selected design point, while the **Metric Ranges** card summarizes the minimum and maximum values observed during the sweep. This helps the user compare the optimum point with the overall behavior of the design space.

6 Results and Discussion

This section presents selected case studies generated using the developed tool.

6.1 Effect of Boiler Pressure and Turbine Inlet Temperature

The first case study considers a relatively intuitive sweep in which boiler pressure and turbine inlet temperature are varied while condenser pressure is kept constant. The purpose of this study is to illustrate how realistic engineering constraints affect the optimal design of the Rankine cycle.

Table 6.1.1: Input parameters used for the boiler pressure and turbine inlet temperature sweep.

Parameter	Value
Sweep variables	Boiler pressure, turbine inlet temperature
Fixed variable	Condenser pressure = 10 kPa
Boiler pressure sweep	8 – 20 MPa, step = 0.1 MPa
Turbine inlet temperature sweep	400 – 530 °C, step = 1 °C
Pump isentropic efficiency	85%
Turbine isentropic efficiency	90%
Objective function	Maximize thermal efficiency
Constraints	Not applied

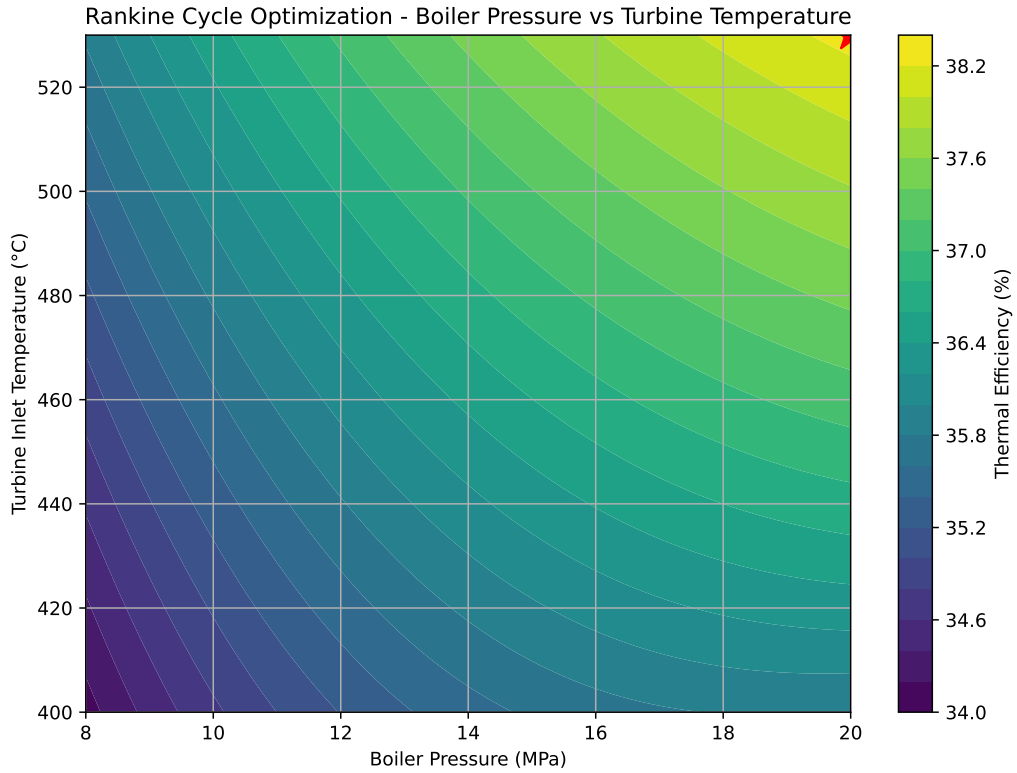


Figure 6.1.1: Unconstrained optimization map for the boiler pressure and turbine inlet temperature sweep.

As shown in Figure 6.1.1, the maximum thermal efficiency occurs at the highest boiler pressure and highest turbine inlet temperature within the selected sweep range. This behavior is expected, since increasing the boiler pressure raises the average temperature of heat addition, while increasing the turbine inlet temperature increases the degree of superheat and generally improves cycle performance.

However, the unconstrained optimum does not necessarily represent a practically feasible design. In a real-world application, additional engineering considerations must be taken into account, such as maintaining a sufficient turbine outlet quality and limiting the required heat input.

Table 6.1.2: Constraints applied in the constrained case.

Constraint	Value
Minimum turbine outlet quality	$x_4 \geq 0.85$
Minimum net work output	Not applied
Maximum heat input	$q_{in} \leq 3180 \text{ kJ/kg}$
Minimum thermal efficiency	Not applied

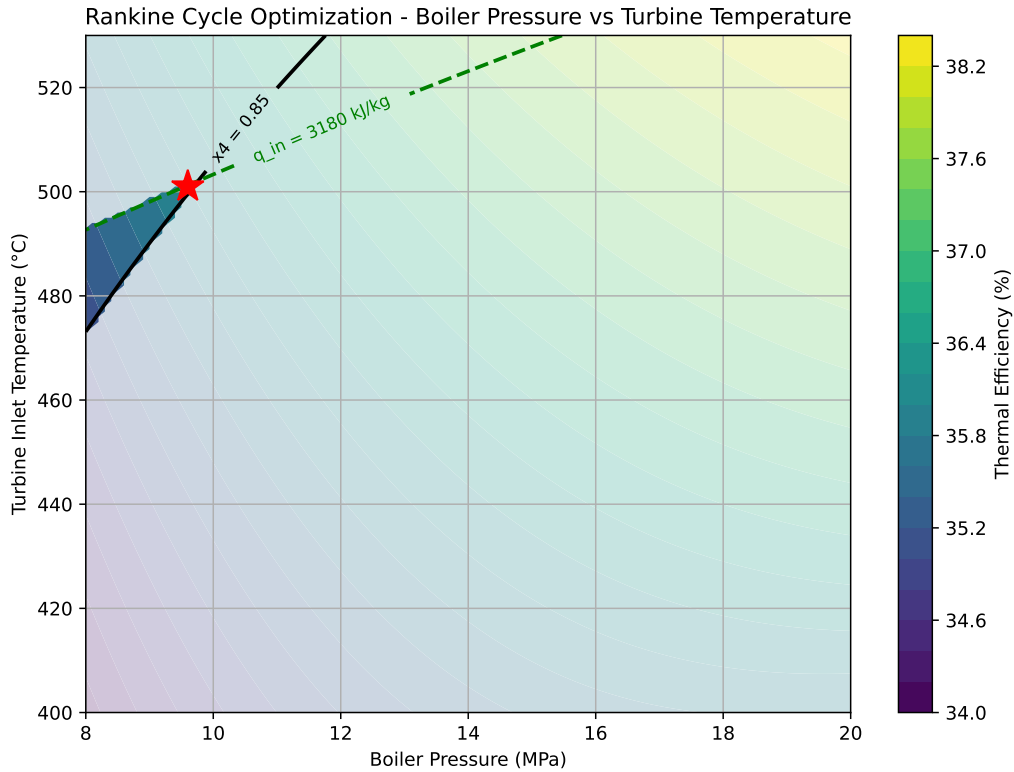


Figure 6.1.2: Constrained optimization map for the boiler pressure and turbine inlet temperature sweep.

Figure 6.1.2 shows the same sweep after the selected engineering constraints are applied. It can be seen that the feasible design region becomes narrower and the optimum point shifts accordingly. In this case, both the turbine outlet quality constraint and the heat input constraint play important roles in determining the feasible maximum-efficiency point.

The turbine outlet quality constraint excludes operating points with excessive moisture content at the turbine exit, while the heat input constraint removes designs that require a larger energy input than the selected limit. As a result, the constrained optimum is selected from a smaller and more realistic design region rather than from the entire thermodynamic sweep domain. This result shows that the best thermodynamic point is not always the most realistic engineering design.

Table 6.1.3: Comparison of unconstrained and constrained optimum points.

Parameter	Unconstrained optimum	Constrained optimum
Boiler pressure	20 MPa	9.60 MPa
Turbine inlet temperature	530.00 °C	501.00 °C
Thermal efficiency	38.26 %	36.00 %
Net work	1193.98 kJ/kg	1144.48 kJ/kg
Heat input	3120.50 kJ/kg	3179.42 kJ/kg
Turbine outlet quality	0.805	0.8507

6.2 Effect of Condenser Pressure and Boiler Pressure

The second case study investigates the effect of varying condenser pressure and boiler pressure while keeping the turbine inlet temperature constant. This sweep is used to examine how the pressure limits of the Rankine cycle affect thermal efficiency, turbine outlet quality, and the location of the optimum design point.

Three comparisons are made in this case study. First, the pressure sweep is performed at a relatively low turbine inlet temperature to observe whether the maximum efficiency point follows the expected high-boiler-pressure and low-condenser-pressure trend. Second, the same sweep is repeated at a higher turbine inlet temperature to examine how additional superheating changes the efficiency distribution and the optimum pressure combination. Finally, engineering constraints are applied to the high-temperature case in order to observe how feasibility limitations modify the available design region and shift the optimum point.

Through these comparisons, this case study demonstrates that the optimum pressure combination is influenced not only by the pressure limits themselves, but also by turbine inlet temperature and practical engineering constraints.

First, a lower turbine inlet temperature is considered in order to observe the pressure-dependent behavior of the cycle under limited superheating.

Table 6.2.1: Input parameters used for the condenser pressure and boiler pressure sweep at low turbine inlet temperature.

Parameter	Value
Sweep variables	Condenser pressure, boiler pressure
Fixed variable	Turbine inlet temperature = 400 °C
Condenser pressure sweep	10 – 22 kPa, step = 0.25 kPa
Boiler pressure sweep	8 – 21 MPa, step = 0.25 MPa
Pump isentropic efficiency	85%
Turbine isentropic efficiency	90%
Objective function	Maximize thermal efficiency
Constraints	Not applied

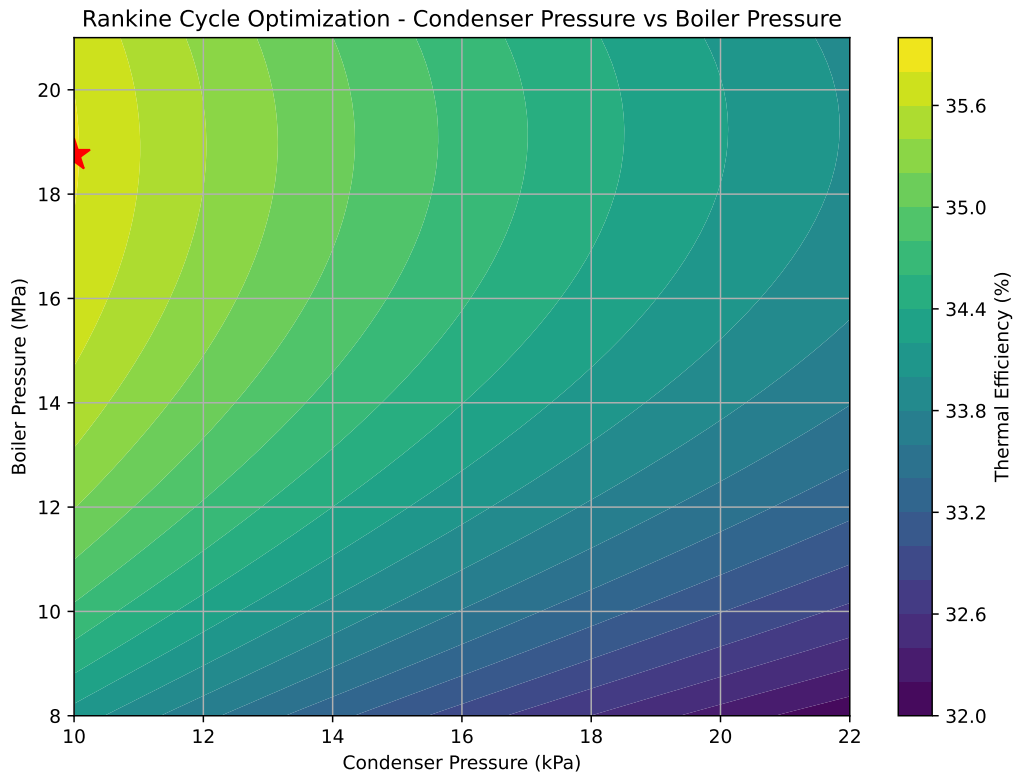


Figure 6.2.1: Unconstrained optimization map for the condenser pressure and boiler pressure sweep at low turbine inlet temperature.

As shown in Figure 6.2.1, the maximum thermal efficiency does not necessarily occur at the highest boiler pressure and lowest condenser pressure within the selected design domain. This indicates that, at lower turbine inlet temperatures, increasing the pressure ratio does not always lead to a direct improvement in thermal efficiency. The benefit of increasing boiler pressure becomes limited when the turbine inlet state is not sufficiently superheated.

Next, the same pressure sweep is repeated at a higher turbine inlet temperature.

Table 6.2.2: Input parameters used for the condenser pressure and boiler pressure sweep at high turbine inlet temperature.

Parameter	Value
Sweep variables	Condenser pressure, boiler pressure
Fixed variable	Turbine inlet temperature = 530 °C
Condenser pressure sweep	10 – 22 kPa, step = 0.25 kPa
Boiler pressure sweep	8 – 21 MPa, step = 0.25 MPa
Pump isentropic efficiency	85%
Turbine isentropic efficiency	90%
Objective function	Maximize thermal efficiency
Constraints	Not applied

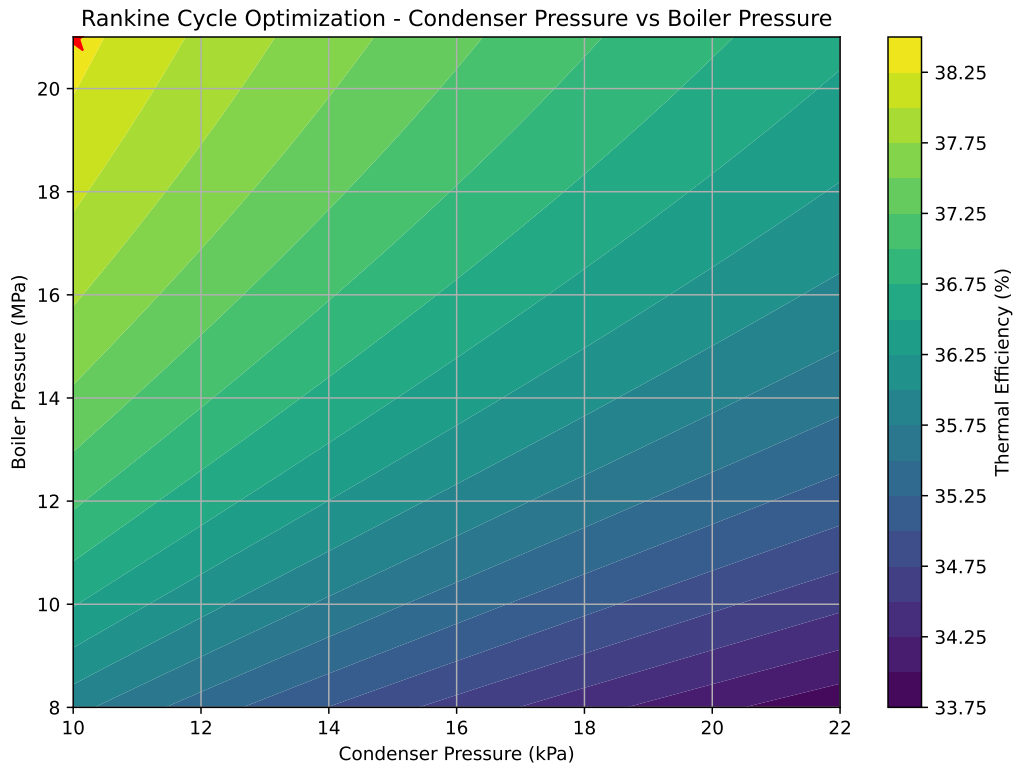


Figure 6.2.2: Unconstrained optimization map for the condenser pressure and boiler pressure sweep at high turbine inlet temperature.

Figure 6.2.2 shows that increasing the turbine inlet temperature changes the location of the optimum point. At a higher turbine inlet temperature, the cycle can benefit more effectively from high boiler pressure and low condenser pressure. This comparison demonstrates that the pressure combination giving the highest efficiency depends strongly on the turbine inlet temperature.

Finally, engineering constraints are applied to the high-temperature pressure sweep in order to examine how practical feasibility limits affect the optimum design point. In this constrained case, the objective function is still selected as maximizing thermal efficiency, but the optimum point is chosen only among the valid design points that satisfy the

imposed constraints. A smaller grid spacing is used in this case to improve the resolution of the constrained design region. Since the feasible region becomes narrower after the constraints are applied, a finer grid allows the constraint boundaries and the constrained optimum point to be represented more accurately.

Table 6.2.3: Input parameters used for the constrained condenser pressure and boiler pressure sweep at high turbine inlet temperature.

Parameter	Value
Sweep variables	Condenser pressure, boiler pressure
Fixed variable	Turbine inlet temperature = 530 °C
Condenser pressure sweep	10 – 22 kPa, step = 0.1 kPa
Boiler pressure sweep	8 – 21 MPa, step = 0.1 MPa
Pump isentropic efficiency	85%
Turbine isentropic efficiency	90%
Objective function	Maximize thermal efficiency
Constraints	Applied, see Table 6.2.4

Table 6.2.4: Constraints applied in the constrained case.

Constraint	Value
Minimum turbine outlet quality	$x_4 \geq 0.82$
Minimum net work output	$w_{net} \geq 1150 \text{ kJ/kg}$
Maximum heat input	$q_{in} \leq 3200 \text{ kJ/kg}$
Minimum thermal efficiency	$\eta_{th} \geq 37\%$

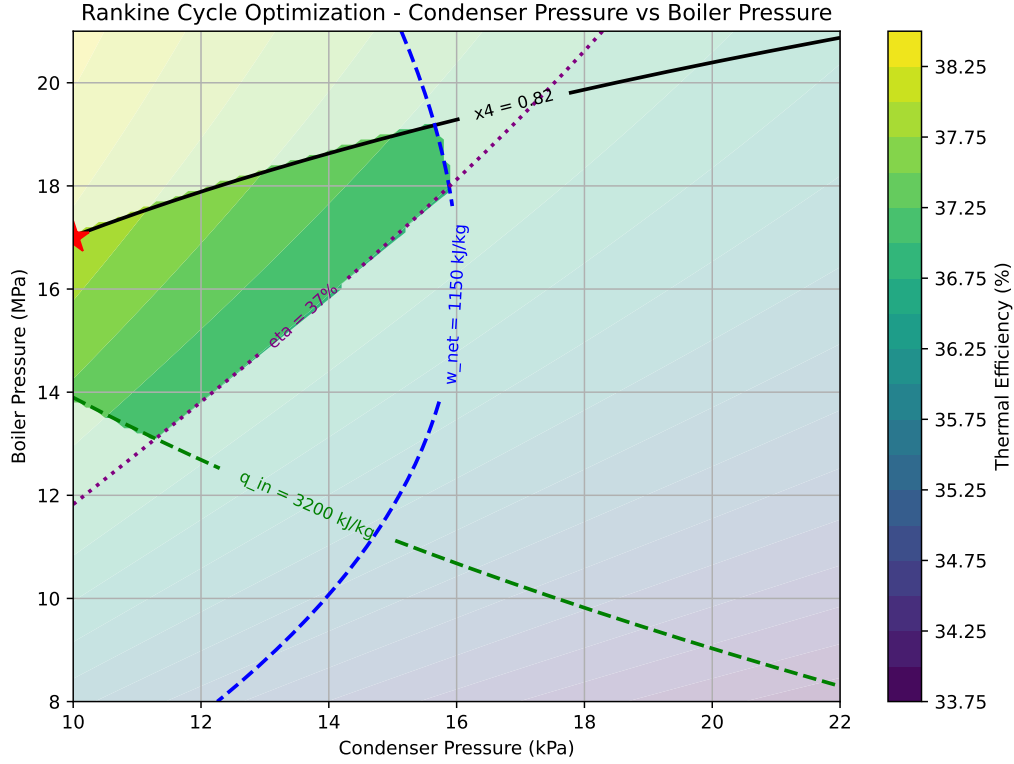


Figure 6.2.3: Constrained optimization map for the condenser pressure and boiler pressure sweep at high turbine inlet temperature.

In Figure 6.2.3, the engineering constraints are applied to the high-temperature pressure sweep while the objective function remains maximizing thermal efficiency. The red point represents the highest-efficiency design among the feasible operating points. This is different from the unconstrained optimum, because some high-efficiency pressure combinations violate the imposed constraints, especially the turbine outlet quality limit. Unlike the first case study, where both turbine outlet quality and heat input constraints influenced the feasible optimum, the turbine outlet quality constraint is more dominant in this pressure sweep. This is because high boiler pressure combined with low condenser pressure increases the turbine expansion range, which can move the turbine outlet state deeper into the wet region. The heat input constraint also removes part of the design domain, but its effect is less restrictive compared with the quality constraint. As a result, the constrained optimum shifts away from the purely thermodynamic maximum and moves toward a more realistic operating region.

Table 6.2.5: Comparison of optimum points for the condenser pressure and boiler pressure sweep.

Parameter	Low-temperature optimum	High-temperature optimum	Constrained optimum
Turbine inlet temperature	400.00 °C	530.00 °C	530.00 °C
Boiler pressure	18.75 MPa	21.00 MPa	17.00 MPa
Condenser pressure	10.00 kPa	10.00 kPa	10.00 kPa
Thermal efficiency	35.82%	38.35%	37.92%
Net work	948.21 kJ/kg	1191.62 kJ/kg	1198.46 kJ/kg
Heat input	2647.35 kJ/kg	3106.97 kJ/kg	3160.23 kJ/kg
Turbine outlet quality	0.710331	0.800716	0.820118

Overall, the condenser pressure and boiler pressure sweep shows that the optimum pressure combination is affected by both turbine inlet temperature and engineering constraints. The results demonstrate that Rankine cycle optimization cannot always be reduced to choosing the highest boiler pressure and lowest condenser pressure. At lower turbine inlet temperatures, the maximum-efficiency point may move away from the extreme pressure-ratio corner, while at higher turbine inlet temperatures the expected high-pressure-ratio trend becomes more dominant. However, when engineering constraints are applied, the feasible design region becomes narrower and the optimum point is selected from the remaining valid operating points.

Therefore, the best pressure combination depends on the combined effects of pressure levels, turbine inlet temperature, and practical feasibility limits.

6.3 Objective Function Comparison

The same constrained high-temperature pressure sweep is used to investigate the effect of objective function selection. The sweep ranges, turbine inlet temperature, and constraints are kept unchanged so that only the selected objective function is varied.

For the same feasible design space, maximizing thermal efficiency, maximizing net work output, minimizing heat input, and maximizing turbine outlet quality may lead to different optimum operating points. Although the contour map displays thermal efficiency, each marker represents the optimum point according to its corresponding objective function. Therefore, the most efficient design is not necessarily the best design when another performance target is prioritized.

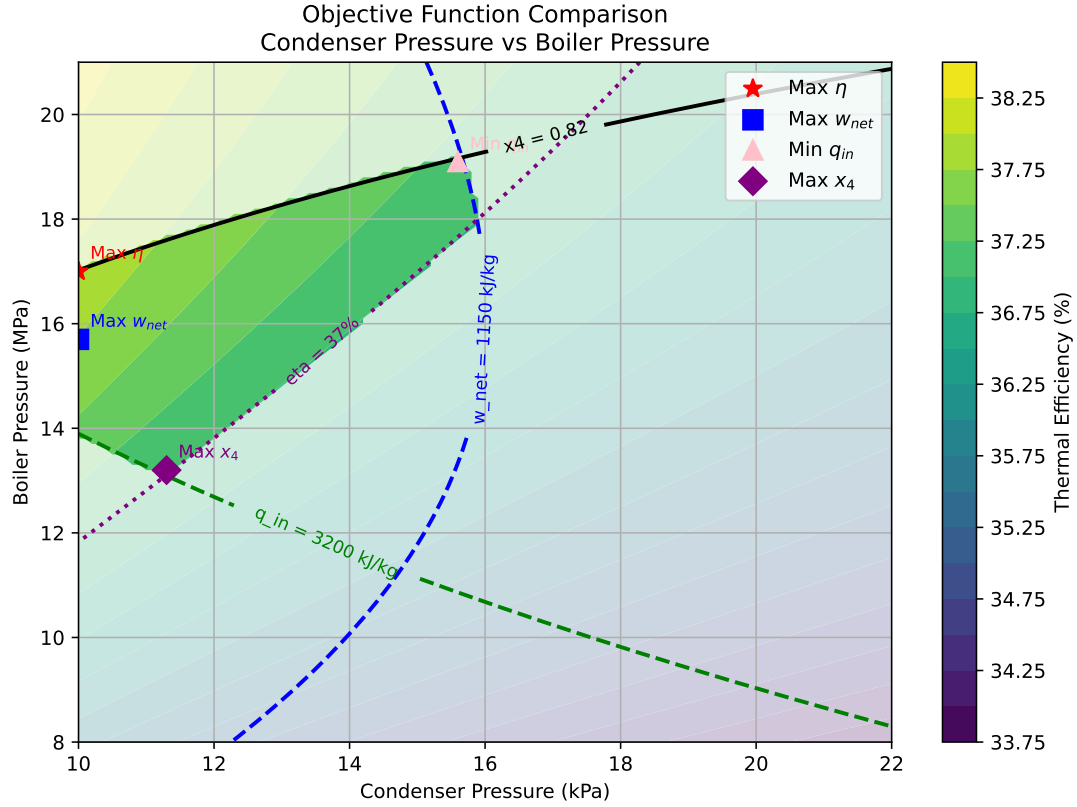


Figure 6.3.1: Comparison of optimum design points for different objective functions on the same constrained optimization map. The contour map shows thermal efficiency, while the markers indicate the optimum points selected according to different objective functions.

Table 6.3.1: Comparison of optimum points for different objective functions.

Parameter	Max. η_{th}	Max. w_{net}	Min. q_{in}	Max. x_4
Boiler pressure	17.00 MPa	15.70 MPa	19.10 MPa	13.20 MPa
Condenser pressure	10.00 kPa	10.00 kPa	15.60 kPa	11.30 kPa
Thermal efficiency	37.92%	37.74%	37.17%	37.02%
Net work	1198.46 kJ/kg	1198.97 kJ/kg	1150.51 kJ/kg	1184.10 kJ/kg
Heat input	3160.23 kJ/kg	3177.05 kJ/kg	3094.92 kJ/kg	3198.64 kJ/kg
Turbine outlet quality	0.820118	0.826938	0.820301	0.844228

As shown in Table 6.3.1, different objective functions lead to different optimum operating points within the same constrained design space. For example, minimizing heat input reduces the required heat input from 3160.23 kJ/kg to 3094.92 kJ/kg, while the thermal efficiency decreases only from 37.92% to 37.17%. This shows that a lower heat input requirement can be achieved with a relatively small efficiency penalty.

Maximizing turbine outlet quality increases the outlet quality from 0.820118 to 0.844228, while the thermal efficiency decreases from 37.92% to 37.02% and the heat input increases from 3160.23 kJ/kg to 3198.64 kJ/kg. This indicates that a safer turbine outlet condition can be obtained, but at the cost of lower thermal efficiency and higher heat input requirement.

Overall, this comparison shows that Rankine cycle optimization involves trade-offs between efficiency, work output, heat input, and turbine outlet quality. There is not a single universally best design point; instead, the most appropriate operating condition depends on the user’s engineering judgment and design priorities.

7 Conclusion and Further Developments

7.1 Conclusion

In this project, an interactive Rankine cycle optimization tool was developed using Python. The tool evaluates a simple vapor power cycle over a two-variable design grid, calculates the main performance parameters, applies engineering constraints, and identifies the optimum feasible operating point according to a user-defined objective function. The graphical interface allows these inputs to be modified without changing the source code.

The case studies show that the optimum operating point is strongly affected by feasibility constraints and the selected objective function. Although higher boiler pressure and turbine inlet temperature can improve efficiency, practical constraints such as turbine outlet quality and heat input limits can shift the feasible optimum point. The condenser pressure and boiler pressure sweep also showed that the optimum pressure combination depends on turbine inlet temperature and engineering constraints.

The objective function comparison demonstrated that different design priorities can lead to different optimum operating points within the same feasible design space. Therefore, the most appropriate operating condition depends on the user’s engineering priorities, such as efficiency, net work output, heat input requirement, or turbine outlet quality.

7.2 Further Developments

Several improvements can be made in future versions of the tool. The current model is limited to a simple Rankine cycle, so reheat and regenerative feedwater heating may be added to represent more realistic steam power plants. Future versions may also include exergy analysis to identify the major sources of irreversibility in the cycle. In addition, economic parameters such as fuel cost, equipment cost, or cost per unit net work can be introduced to connect thermodynamic optimization with practical design decisions. More advanced optimization methods may also reduce the computational cost of grid-based sweeps and locate optimum points more efficiently.

Overall, the developed framework helps visualize Rankine cycle performance, interpret thermodynamic trade-offs, and select feasible design points under engineering constraints. It can also serve as a foundation for more advanced cycle analysis and optimization studies.